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Design and implementation of a semantic gateway based on SSN ontology

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Abstract

The massive data in the industrial big data field mainly comes from a large number of sensors. Sensor data is discrete numerical data with heterogeneity. In scenarios such as edge computing, it is often necessary to mine the relationships between the data in the data stream so as to realize the full use of heterogeneous sensor data. Because the sensor data does not contain contextual semantic relations, the relationship between the data is difficult to be discovered. Aiming at the above shortcomings, we design and implement a semantic gateway with semantic annotation function. The original sensor data will become RDF data with semantic information after being processed by the semantic gateway. Our work is mainly divided into two aspects: according to the source and actual characteristics of the data, we choose to carry out the semantic work based on the SSN ontology; next, we propose an OB2R algorithm to realize the semantic annotation function of the data, and the correctness of the algorithm is verified on the PHM2010 dataset.

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Keywords: industrial big data; SSN ontology; semantic annotation

1. Introduction

Industrial big data is a large amount of data generated by industrial equipment, corresponding to the state of the equipment at different times¹, and it is the information on the Internet of Things². Its origin can be roughly divided into three aspects. The first is production and operation, which is the necessary data for enterprises. The second is

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the interconnection of devices. People began to deploy a large number of sensors in different scenarios to monitor equipment or the surrounding environment. In this case, sensors began to become the main source of industrial big data. The third is external data, which mainly includes other data generated by corporate activities, such as data related to market launches. The industrial big data discussed in this paper defaults to sensor data generated by the interconnection of devices. Semantic annotation has many application scenarios in daily life, such as smart home, environmental monitoring, and security alarms. Therefore, how to perform semantic annotation on these isolated sensor data, give the semantic relationship between the data, mine the association between the data, and find useful information through data reasoning is a problem worthy of study.

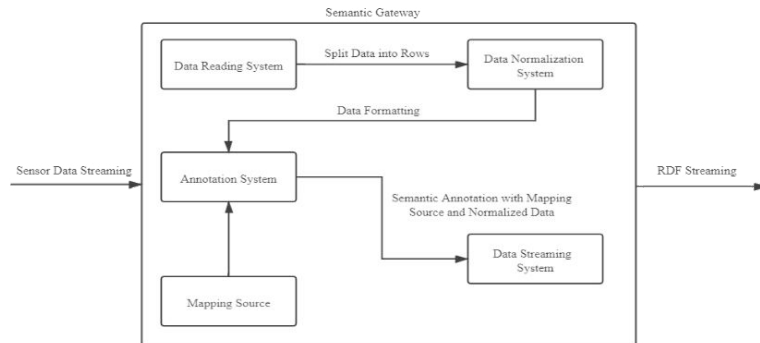


Fig. 1. Internal Structure of the Semantic Gateway

Fig.1. shows the internal structure of the Semantic Gateway. The input of the semantic gateway is the sensor data stream, and the output is the RDF stream containing semantic information. A semantic gateway should include a data reading system for receiving original data, a data normalization system for converting the original data format, an annotation system for annotating data based on mapping resources or corresponding relationships; a data fluidization system for data to the data stream. Fig.2. shows a schematic diagram of the process of integrating and processing data in scenarios such as edge computing and the location of the semantic gateway designed in this article.

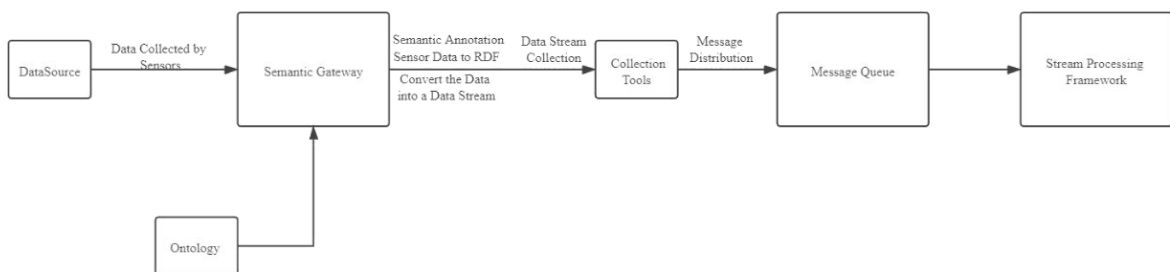


Fig. 2. Overall Structure of System

As shown in Fig.2., after the sensor data is annotated by the semantic gateway and streamed into an RDF stream, use a data collection tool such as Flume to collect the RDF stream and distribute it to the message queue. We can use the stream processing framework to pull these data from the message queue for consumption. For example, we can use Flink to consume data in Kafka. Flink uses a window mechanism to construct an RDF graph in real-time, and users can perform online SPARQL queries on the RDF graph in a fixed window. You can also store the RDF data for offline calculation, and use it for equipment operation and production process monitoring.

2. Related Work

There are professional ontologies constructed by professionals in various fields. For example, partial ontologies in the field of geographic information are shown as follows: the civic sensor project of FaceBook⁵, SMEAR⁶, LinkedIn Data^{7,8}, Got^{9,10} ontology of Berlin project, and SMO¹¹ontology. For industrial big data scenarios, since the data mainly comes from sensors, the SSN ontology can be used.

Sensor data can be manually converted into RDF that conforms to the SSN ontology specification, but given the enormous amount of data generated in real-world industrial production scenarios, the manual conversion is not feasible.

In addition, there are some annotation methods based on mapping relationships, such as D2RQ¹², R2RML¹³. These methods usually need to establish a mapping file, provide a corresponding relationship through the mapping file, and map the original data to the corresponding position of the ontology. Although this method does not require us to operate manually, the configuration file needs to be defined in a specific scenario, resulting in poor versatility. Semantic annotation based on SSN or extended ontology has been applied in many fields. For example, Gu Jiye et al. from Wuhan University proposed the rainstorm and flood disaster chain ontology by expanding SSN¹⁴. Zhao Yunping and Li Hongwei et al. from Hebei University of Science and Technology conducted a series of relevant studies and optimization based on SSN for the application scenarios of vegetable greenhouse temperature monitoring and national air quality monitoring^{15,16}. In order to overcome the difficulty that SSN contains less geo-related information, Ai Qing et al. from Hebei University of Science and Technology extended SSN ontology and applied it to the query of geographic information¹⁷. On the basis of previous work, in view of the source and characteristics of industrial big data, we decided to use the SSN ontology to model the data, and use the ontology analysis function of Apache Jena to analyze the SSN ontology, and assign the sensor data to the corresponding Then, Apache Jena writes out the assigned ontology to a file to realize the semantic annotation function of the sensor.

3. Semantic Gateway Design and Implementation

3.1. Ontology Analysis

As the data preprocessing module, the semantic gateway is used to realize the semantic annotation of data directly based on the SSN ontology. Further, considering that our use scenario mainly involves observing the data collected by the sensor, we decided to mainly use the Observation module in the SSN ontology for ontology construction. The SSN ontology classes and related attributes and their relationships involved in this article are shown in Fig.3.

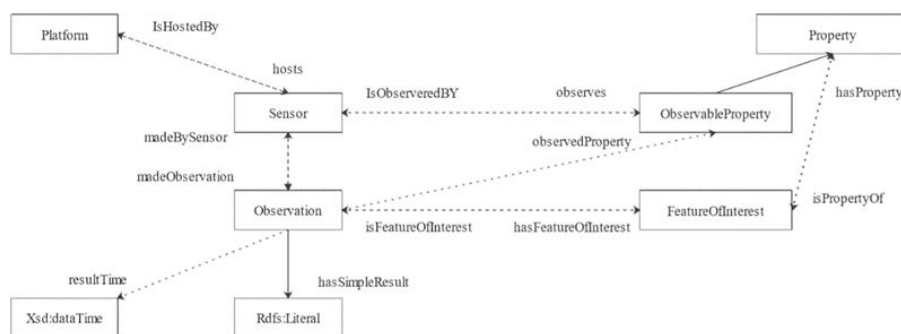


Fig. 3. SSN Ontology Classes Properties and Their Relationships in This Paper

After completing the ontology selection, the core system design of the semantic gateway needs to be carried out. First, we need to prepare the mapping resources. We set up a properties configuration file to store the correspondence between each sensor and the id number. Then, we set up an XML mapping file to store the detailed information of each sensor device. The content of the XML mapping file corresponds to the content of the properties configuration file. DOM4J can be used to parse the XML mapping file to get the device id contained in the tag body, use device id find the corresponding device name in the properties file. The data normalization system of the gateway is designed as follows: each dimension of data is separated by a comma, and each group of data is separated by a semicolon; then the corresponding sensor information is added to each group of data through the above mapping resources, to facilitate subsequent processing. About the annotation system of the semantic gateway, we propose an annotation algorithm OB2R (Observation to RDF) from normalized data to RDF data(see Table1.).

3.2. OB2R Algorithm

Table 1.The specific process of OB2R.

Algorithm -OB2R
Input:
the sensor data stream;
the XML mapping file;
the properties file;
the SSN ontology file;
Output:
SSN ontology file with instances;
Steps:
1. read properties file,get ID and corresponding name instances;
2. create OntologyModel based on SSN ontology file;
3. parse ontology file,create Platform and FeatureOfInterest OntClass;
4. parse ontology file,create Sensor,Observation and ObservableProperty OntClass ;
5. create individuals based on ID instance;
6. parse ontology file,create OntProperty of each OntClass;
7. assign each OntProperty to the corresponding OntClass;
8. while(dataStream != null){
9. sensorDataArr = dataStream.split(";");
10. for(sensorData data: sensorDataArr){
11. sensorPropertyArr = data.split(",");
12. create individual I _{observation} of Observation;
13. for(sensorProperty property :sensorPropertyArr){
14. read I _{sensor} 、I _{property} and I _{featureOfInterest} from property, assign them to the corresponding attribute;
15. }
16. assign attributes,hasSimpleResult and resultTime to I _{observation} ;
17. }
18. }
19. persist result as a owl file;

As shown in Table 1, the algorithm is to semantic the normalized data. Firstly, we read the properties file to get the instance name and ID(line 1). Next, read the SSN ontology file and obtain the OntModel (line 2). Use Ontmodel.getontClass () to get the Platform class, Sensor class, Observation class, FeatureOfInterest class, and ObservableProperty class (lines 3-4). Create the corresponding instance according to the ID in the Properties file. Create an instance of the corresponding class with the instance number prefix (line 5). Get the object properties using the Ontmodel.getontProperty () (lines 6-7). For each piece of data transferred to the semantic gateway, a semicolon is used to separate the array of individual sensor data (line 9), and for each piece of data in the array, a comma is used to separate the field values for each piece of data (line 11). Using Observation. CreateIndividual () to create instance I_{observation}(line 12). I_{sensor}, I_{property}, and I_{featureOfInterest} were obtained from the URL of the instance, and associated with I_{observation} through hasFeatureOfInterest, observedProperty, madeBySensor (line 14). Finally, use hasSimpleResult and resultTime to correlate the observations and times (line 16). When the semantic annotation is complete, the result is persisted in the OWL file in the specified format (line 19).

3.3. Experimental Analysis

In order to verify the feasibility of the algorithm, semantic annotation experiments were carried out on the PHM 2010 dataset. Experimental conditions are as follows:

The experimental equipment includes a Roders Tech RFM760 CNC milling machine, a 3-gear ball head carbide milling cutter, a rectangular Inconel 718 milling material, a Kistler5019A multi-channel charge amplifier, a Kistler 9265B three-way dynamometer, a NI DAQ data acquisition card, and a LEICA MZ12 microscope.

Operation parameters of Roders Tech RFM760 NC machine tool are as follows:

Tool spindle speed is 10400rpm; The feeding rate in the X-direction is 1555mm/min. The depth of radial cutting in Y-direction is 0.125mm; The axial cutting depth in the Z-direction is 0.2mm. The 3-way platform dynamometer is installed between the workpiece and the processing platform, the 3-way vibration sensor is fixed on the processing platform, and the ACOUSTIC emission sensor is installed near the workpiece. The electric charge amplifier is used to amplify the signal, and the data acquisition card is used to collect the cutting force, vibration, and acoustic emission signals in three directions during the processing. The sampling frequency of each signal is 50kHz. Each record of the collected data can be divided into seven dimensions: forces in the direction of X, Y and Z; vibration in the direction of X, Y and Z, and acoustic emission signals (see Table 2).

The sample data structure collected by the sensor is shown in Table 3.

Table 2. Properties in the Collected Data.

Number	Property Name
1	Force (N) in X dimension
2	Force (N) in Y dimension
3	Force (N) in Z dimension
4	Vibration (g) in X dimension
5	Vibration (g) in Y dimension
6	Vibration (g) in Z dimension
7	AE-RMS (V)

Table 3. Sample Data Collected by Sensors

XForce	YForce	ZForce	XVibration	YVibration	ZVibration	AERMS
-0.22	-0.363	1.925	-0.048	-0.073	-0.051	-0.005

The XML mapping file fragment is as follows:

```
<sensor id = "1">
  <platform>pl-0001</platform>
  <name>se-0001</name>
  <featureOfInterest>fe-0001</featureOfInterest>
  <property>pr-0001</property>
</sensor>
```

As shown in Table 4, the id prefix of pl represents the Platform instance, se represents the Sensor instance, fe represents the FeatureOfInterest instance, and pr represents the ObservableProperty instance. The instance number is unique. In the process of analyzing the sensor, we first define a Sensor class to store sensor information, and then parse the XML mapping file through DOM4J, and go to the properties configuration file to obtain the sensor information according to the id value of each Sensor contained in the <name> tag, and store it in List<Sensor>, get a list of sensor instances. For each acquisition of the real-time data stream generated by a group of sensors, we need to normalize the original data, that is, split a group of data containing multiple sensor observations into multiple groups of data corresponding to a single sensor, and the field of the single sensor data is data, We expand the data of a single sensor, and use the sensor information in List<Sensor> to add the sensorID, observedPropertyID,

featureOfInterestID, and time fields of the corresponding sensor. The data in a single sensor is separated by a comma, and the data between a group of sensors is separated by a semicolon. After the processing is completed, it is handed over to the semantic gateway for processing. The normalized data is shown in Table 5.

Table 4. Example of Properties Configuration File

Instance Name	Instance Number	Instance Meaning
CNCMachineTools	pl-0001	CNCMachine Platform
xDynamometer	se-0001	Dynamometer in X Direction
yDynamometer	se-0002	Dynamometer in Y Direction
zDynamometer	se-0003	Dynamometer in Z Direction
xAccelerometer	se-0004	Accelerometer in X Direction
yAccelerometer	se-0005	Accelerometer in Y Direction
zAccelerometer	se-0006	Accelerometer in Z Direction
acousticEmissionSensor	se-0007	AcousticEmissionSensor
tool	fe-0001	Tool of CNCMachine
xForce	pr-0001	Milling Forces in the Direction of X
yForce	pr-0002	Milling Forces in the Direction of Y
zForce	pr-0003	Milling Forces in the Direction of Z
xVibration	pr-0004	Vibration in the Direction of X
yVibration	pr-0005	Vibration in the Direction of Y
zVibration	pr-0006	Vibration in the Direction of Z
AE-RMS	pr-0007	Acoustic Emission Signals

Table 5. Normalized Sensor Data

Sensor Number	Data
1	se-0001,pr-0001,fe-0001,0.704,1601360560890
2	se-0002,pr-0002,fe-0001,-0.387,1601360560890
3	se-0003,pr-0003,fe-0001,-1.084,1601360560890
4	se-0004,pr-0004,fe-0001,0.018,1601360560890
5	se-0005,pr-0005,fe-0001,0.031,1601360560890
6	se-0006,pr-0006,fe-0001,0.027,1601360560890
7	se-0007,pr-0007,fe-0001,-0.004,1601360560890

We performed a simple semantic annotation experiment on the normalized data, and the generated data fragments are as follows:

```
sosa:se-0003_pr-0003_fe-0001_1601360584671
  a
  sosa:hasFeatureOfInterest sosa:fe-0001 ;
  sosa:hasSimpleResult      "1.88" ;
  sosa:madeBySensor         sosa:se-0003 ;
  sosa:observedProperty     sosa:pr-0003 ;
  sosa:resultTime           "1601360584671" .
```

In order to verify the correctness of the generated data, we check the saved OWL file by manual verification to ensure that there are no obvious errors. After the manual verification is completed, open the annotation result in Protégé, a series of data instances appear in Observation, indicating that Protégé can recognize the results of our semantic annotation.

4. Summary and Outlook

We started with the characteristics of sensor data and SSN ontology, realized the expected semantic annotation function based on SSN ontology, and completed the design and implementation of the semantic gateway. This semantic annotation method is universal. If you switch the semantic annotation scene, you only need to rewrite the XML mapping file and the properties configuration file. Taking into account the versatility and project integrity, a graphical interface for scene configuration will be developed in the future. The XML mapping file and the properties configuration file can be automatically generated by configuring on the operation interface, without the need to create files manually. Of course, the semantic gateway designed in this article only implements the most basic semantic annotation function, and there are many aspects that can be improved. For example, the number of data streams connected by the semantic gateway and the ability of data distribution after streaming is further optimized; the support of the semantic gateway for mainstream open ontology is increased. These all need follow-up research and improvement.

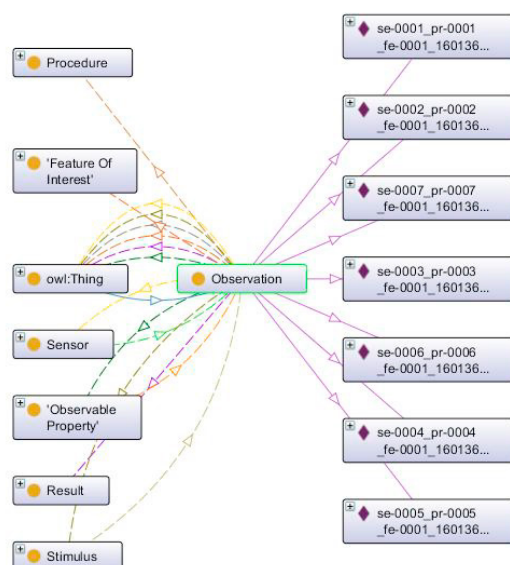


Fig. 4. Annotation Result in Protégé

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